Images on Web Pages

- Only an Intro
- A lot of concepts
- A lot of potential interactions

Images are separate (usually)

- HTML
- CSS
- Images as separate files
    - SOME small images are "inline"

Images come in many formats

- All are collection of info
    - Binary values
    - Format defines how that binary is interpreted
- Common formats:
    - jpg/jpeg, gif, png, svg. .mp4, .webp
- Many more formats exist
    - Limited browser support

Images have many uses

- Used as foreground "objects"
    - Such as text, buttons, etc
    - These fit in document flow
    - These are "content"
- Used as backgrounds
    - Like background colors
    - These are *background*
- Used as icons
    - Smaller foreground objects
    - Not all icons are images
- Favicon (used in browser tabs)

The `img` element

- Empty/void/self-closing element
- url for image is `src` attribute
- `alt` attribute should always be used
    - With meaningful, accurate description!
    - UNLESS image is purely for visual impact
        - Ex: a floral decoration on a side of a page
        - THEN have `alt=""`
        - But should be background, not content

A Figure

> *an image, illustration, diagram, code snippet, etc., that is referenced in the main flow of a document, but that can be moved to another part of the document or to an appendix without affecting the main flow*

Not all images are figures, but some are

- A lot of inconsistency about this

Not all figures are images, but most are

Figure Element

A `<figure>` element can wrap an `<img>`

- Optionally, a `<figcaption>` child of the `<figure>` (and sibling of any `<img>`)
    - `<figcaption>` would be any caption text
    - `<figcaption>` - What the picture doesn't say
    - `alt` text - What the picture DOES show you
- May have other child elements
    - Whatever creates the "figure"

Caption vs Alt

- `alt` is the visual of the picture in text
    - What you SEE
- `caption` is info about the picture
    - What the pic doesn't show without context
- Poorly captioned/alt-ed/semantic imgs common!
    - Ignorant of impact, think "good enough"

What's wrong here?

- **https://www.snopes.com/fact-check/margaret-hamilton-moon-landing-code/**
- **https://airandspace.si.edu/stories/editorial/rope-mother-margaret-hamilton**

Styling an image

- `aspect-ratio` sets the width/height ratio
    - Handy when resizing to maintain ratio
    - Avoids "streched" or "squished" images
- Desire image files that match dimensions in use
    - Most people don't resize much
    - Too big/too small is wasteful/slow/blurry
- `vertical-align`
    - `baseline` - allows space below for text like "g"
    - `bottom` - not baseline, no space allowed

Background images

For when **image is not content**

- If a user needs it, don't make it background
- No `alt` text option
- Image loaded by CSS, not by HTML
- CSS is on some HTML element

Background property

- `background-color`, `background-image`
    - All part of `background-` collection
- Transparent image parts will try to show color

`background-image` property

- Pass `url()` with the url inside
    - Fully qualified, absolute, or relative
        - Relative to CSS file, not HTML page!
    - Value can be in single, double, or no quotes
        - Punctuation and spaces cause problems
            - Avoid when you can
            - Use quotes when you can't avoid

Other `background-` properties

- `background-position` - Align container and image
- `background-repeat` - If container larger than image
- `background-size` - To resize image
- `background-attachment` - When container scrolls

See MDN for examples

Stock Photos

Pre-existing images

- People, places, things
- Created/Collected in bulk by vendors
    - Example: getty

Using Stock Photos

- Hero Images
- In cards/panels

I'm a poor student!

- **https://unsplash.com/** is a good, free source
- "free" is complicated

Copyright

I am not a lawyer

- I am warning of *potential* risks

Copyright not Universally the same

Berne Convention

- Most countries signed
- Defines common ground
    - Lots of details left to countries

Copyright grants "rights" to creators

- Duplication ("right to copy")
- Derivative Works
- Expression, not Idea
- NOT based on "profit"
- Automatic!

LLM and "AI" use in training data isn't decided(!)

Copyright protects duplication/distribution

- You might download it without cost
- Does not mean you can DO things with it
    - Legally speaking

Wait, are memes illegal?

- Technically…many of them, probably
    - Civil, not criminal
- What you get away with vs what is allowed

When money gets involved people get more serious

- On the job, money is always involved

Derivative Works

- If you change it for your uses
    - That's a "derivative work"
- Translations? Derivative work

Where is the line?

- Lawyers earn their pay

Copyright protects Expression

- Not ideas, not facts
- Not methods
    - That's Patents
- Not identity (name or appearance)
    - That's Trademarks
- Code is usually expression

Lack of profit is no defense

- Copyright is creator's right
    - So it doesn't matter if YOU are making money
    - About if it hurts THEM
        - Profit or profit potential
    - Disagreements resolved in court ($$$)
- Being sued sucks
    - You lose even if you win
- Usually Cease and Desist first
    - C&D may not be legally correct

Copyright is automatic!

- Work protected as soon as created
    - Owner decided in advance
        - Example: "Work for hire"
- Does not require "registration"
    - Registration may help prove timing

"Fair Use" is a defense

- Fair Use
    - Parody
    - Review
    - Education
- When sued, you can claim "fair use" as defense
    - Doesn't mean you win
    - Doesn't stop you from getting sued
- Lots of bad claims of fair use
    - Example: no profit is not fair use

Licenses and Copyright

- Permission to use a copyrighted work
- What is allowed varies by License
- Common:
    - Highly permissive: MIT, BSD, DWTFYW, CC-BY
    - Restricted: CC-BY-NC
    - Copyleft: GPLv2, GPLv3, CC-BY-SA

Beware sites with "Free" downloads

- Do they offer/define a license?

Practical Concerns

- Don't download/use random images/code in work
- Track where something came from
    - And what license it has
- Very different conversations with legal dept
- Know employer's rules about open source
    - Often based on specific licenses

Trademark protects CONSUMER

- Limited identity rights to owner
    - to prevent consumer confusion
- Different kinds (trademark, tradedress, etc)
- "requires owner to protect" (ish)
    - They lose the protection if it becomes generic
    - Kleenex, McDonalds, Xerox, Google

Website Design can be Trade Dress!

- Do not copy a sites' design too literally
- Be "inspired by", not "a copy of"
- Test is whether it confuses users
    - Make your identity clear!
- Hard to know
    - Safest to avoid approaching the line

Icons

Small image-looking things

- Convey info
- May not actually be an image!
- Should have text
    - Either direct or associated

Images?

Icons might be:

- Images
    - SVG allows easy resizing
- Special Font characters
- CSS!

Icons as font example

- Font Awesome is one vendor
    - Offers font-based or SVG-based
    - **https://fontawesome.com/v4/icons/**

https://cdnjs.cloudflare.com/ajax/libs/font-awesome/4.7.0/css/font-awesome.min.css

```
<link rel="stylesheet" href= "https://cdnjs.cloudflare.com/ajax/libs/font-awesome/4.
/* later */
Test <i class="fa fa-mortar-board"></i>
```

Test 🎓

Icons from CSS

- Example: **https://css.gg/**
    - common, simple shapes
    - Provides the CSS to create
        - Or many other formats
    - Used to be MIT License
    - Now? I no longer recommend use
        - But it does show what CSS can do

Icons from Google

- **https://fonts.google.com/icons**
- SVG, PNG, or as a Font
- Apache License

Icons are great

- But remember not to exclude text
    - For a11y text MUST be available
    - Many users will prefer to have anyway

What's with these fonts anyway?

Browsers have Operating System fonts available

- Not all OSes have the same fonts!
- Safe to stick to a very limited set of fonts
- Or load external fonts

Loading a Font

- Loaded via a CSS file
- Then set the font-family

```
<link rel="stylesheet" href="https://fonts.googleapis.com/css2?family=Crimson+Pro">
```

https://fonts.googleapis.com/css2?family=Crimson+Pro

```
body {
  font-family: "Crimson Pro", serif;
}
```

Finding Fonts

- That's a whole thing
- Font people are SERIOUS about fonts
- Google has a widely used, free (CC-BY-SA) collection
- **https://fonts.google.com/**

Deciding on Fonts

https://fonts.google.com/knowledge/choosing_type

- Browser must download custom fonts!
    - Can slow page
    - What do you show before they load?
- Not too many!
    - Page can be very "busy"
- Legible!
    - Fancy Gibberish is still gibberish
- Appropriate Tone
    - **https://comicsanscriminal.com/**